

Physical and mechanical properties of polyvinylidene fluoride - Short sugar palm fiber nanocomposites

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ABSTRACT

The characterization of natural fiber composites has become of paramount importance. The aim of this work is to prepare, produce, and investigate the mechanical, nanomechanical, and physical properties of polyvinylidene fluoride (PVDF) reinforced with short sugar palm fiber (SSPF) composites. Preparation involves successive phases: a melt-mixing approach and a hot/cold compression molding procedure. The mechanical test examines the flexural and tensile strengths. This work introduces nanomechanical testing through the quasistatic nanoindentation to measure the hardness, fracture toughness, stiffness, and Young's modulus. Physical properties are analyzed by measuring the composites' density, water absorption, moisture content, and thickness swelling. Mechanical properties displayed good flexural and tensile enhancements. The use of short fiber led to adequate interfacial bonding between the fiber and matrix. Composites exhibited high compatibility, strength, and a homogenous structure in the fiber-matrix. Nanomechanical test displayed a dynamic modulus mapping on the fiber. The scan line analysis and in-situ scanning probe microscopy (SPM) imaging presented improved hardness, Young's modulus, stiffness, and fracture phenomena. The composites exhibited outstanding physical properties and excellent resistance to water absorption, moisture content, and thickness swelling. The novel findings provide insight on how biocomposite materials can be utilized in various engineering applications. This signifies their high competency and applicability for green production.

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1. Introduction

Fibers are mostly used as reinforcement to yield fiber reinforced composites (FRC) and fiber reinforced polymers (FRP). Numerous drawbacks have been encountered in synthetic fibers which prompted the investigations of other alternatives (Chauhan et al., 2012). The drawbacks include environmental issues, high cost, and high density (Begum and Islam, 2013). Utilization of natural

fibers has noticeably increased in the last two decades due to their distinguished properties. Natural fibers such as hemp, jute, coir, flax, kenaf, sugar palm, sugarcane, oil palm, bamboo, ramie, cotton, silk, wood, abaca, and sisal have been used in polymer composites (Bachtar et al., 2010). The use of natural fiber composites in numerous and advanced applications may open new scientific horizons. They can be substantial alternatives to harmful synthetic materials since they are sustainable resources and environmentally friendly (M.H. Alaaeddin et al., 2019a). Sugar palm fiber (SPF) possess distinct and resilient properties for multiple applications. SPF is known for its high durability and resistance to water. It is currently used as reinforcement in polymer matrix composites (Edhirej et al., 2017). SPF possesses high thermal stability and strength due to the abundant chemical content of cellulose (Huzaifah et al., 2017). Its excellent properties make it suitable for numerous uses and outdoor applications (Jumaidin et al., 2017). SPF is extracted from sugar palm (SP) (*Arenga Pinnata*), a type of palm

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tree found in tropical regions such as Indonesia, Malaysia, Philippines, Thailand, and Southeast Asian countries (Atiqah et al., 2017). It has been employed in several applications such as septic tank base filters, paintbrushes, ropes, chairs, construction materials, etc. (Bachtar et al., 2009). An increasing trend is present regarding their utilization in technological platforms (Alaaeddin et al., 2019b). SPF is a novel source of bio-fibers. Various researches have employed and incorporated SPF in polymer composite structures. Alaaeddin et al. (2018) focused on determining the most appropriate polymer matrix type for short SPF composites. Polyvinylidene fluoride (PVDF) was found to be a suitable candidate in preparing SSPF composites for multi-engineering purposes and outdoor applications. This work introduces a novel PVDF-SSPF composite. Its structure, mechanical, nanomechanical, and physical properties are extensively investigated. SPFs are usually brown or black in color (see Fig. 1) with diameters of up to 0.50 mm (0.020") (Leman et al., 2008a, 2008b). The fiber has several names: black fiber, aren, gomuti, or "ijuk" (as locally known). It is identified as strong, durable, and resistant to water and heat. The fiber exhibits tremendous resistance to sea water and is used in various applications where water is involved. Its naturally woven form allows for easy preparation and processing. Numerous SPF composite products have been successfully developed in the last few years. Additional research is highly encouraged to promote its use in multiple applications (Ishak et al., 2013).

Poor fiber-matrix adhesion have been observed when using certain types of untreated fibers (Colom et al., 2003). Emerging techniques contribute to the strength, adhesion, and preparation of multifunctional composites (Huang et al., 2017). Mussel-inspired methods vastly improve the composites' properties, structure, and bonding. Such procedures are usually recommended for preparing functional, stable, and multilayer films (Zhang et al., 2017).

Natural fiber properties mainly rely on the matrix, fiber type, and interfacial bonding. In fiber-matrix adhesion, efficiency and compatibility are addressed by the stress transfer between reinforcement fibers and the matrix (AL-Oqla et al., 2015). Sahari et al. (2013) found good adhesion between the fiber and matrix. The homogenous distribution improved the mechanical properties of biocomposites. Interfacial adhesion increased with accurate ratios of fiber-matrix and hybridization mechanisms (Atiqah et al., 2018).

SPF is used in numerous composites. It is a unique fiber with excellent properties for many applications (Ishak et al., 2013). It is a strong candidate for thermoplastic composites and reinforcement applications with various fiber loadings. SPFs contribute to favorable mechanical and modulus properties such as impact strength, flexural strength, and flexural modulus. Its cellulose content is high compared to other natural fibers such as kenaf and coir. It has the

potential of being used as reinforcement in natural fiber composites (Mukhtar et al., 2016).

2. Materials and methods

2.1. Materials

Two main materials were acquired to process composites. The first is sugar palm fiber (SPF) extracted from matured sugar palms located at Kuala Jempol, Negeri Sembilan, Malaysia. The second is polyvinylidene fluoride (PVDF) obtained and verified from two different sources in China. One sample was chosen due to certain specifications (see Table 1).

2.2. Preparation of samples

Short sugar palm fiber (SSPF) is prepared through several successive phases.

- (1) The preparatory phase: The fiber threads are manually selected according to length, thickness, and purity in color.
- (2) The crushing phase: Chosen threads of SPF are cut into small fragments with a crushing machine.
- (3) The grinding phase: Crushed specimens undergo another cutting process so as to fit in a sieve analysis machine.
- (4) The sieving phase: Specimens are refined after grinding and undergo a sieve analysis process to obtain 200–250 μm fiber.

Fig. 1 displays SPF before and after processing.

Polyvinylidene fluoride (PVDF) was used in the matrix composite. It is a thermoplastic polymer known for its high impact strength, thermal expansion, hardness, altered density, solubility, and degree of crystallinity. The polymer has better piezoelectric and pyroelectric coefficients. It is a suitable candidate for numerous operations within industrial platforms. PVDF is utilized in several indoor and outdoor applications such as sensing materials, insulations, encapsulations, laminations, membranes, coatings, aircraft interiors, photovoltaic applications, biomaterials, and composite structures (Alaaeddin et al., 2018). Table 1 presents the properties of the employed polymer in matrix composites.

2.3. Fabrication and testing of composites

SSPF and PVDF were fabricated, pressed at different temperatures, and tested. The process was repeatedly performed to ensure the reliability and validity of results.

2.3.1. Fabrication of composites

SSPF and PVDF were combined using the Brabender measuring mixer machine provided by (Brabender, Germany). The machine is specifically designed to realistically simulate the outcome on a laboratory scale. The fiber loading in the matrix was 30%, as recommended in various studies. PVDF was added into the mixer-bowl at a temperature above 175 °C and screw speed of 50 rpm. SSPF was then combined with the melted PVDF. Fig. 2 presents the torque-time plastogram of the blends and their fusion behaviors.

The specimens were pressed by using a compression molding machine. They underwent a preheat of 10 min, a hot press for 5 min, and a cold press for 1 min with maximum pressure of 160 pa. The size of the mold used was 150 mm (L) $\times$ 150 mm (W) $\times$ 2 mm (T). Developed specimens weighed $65.55 \pm \text{g}$. Length and dimension measurements were confirmed to be equal. Fig. 3 presents frontal and cross-sectional views of the obtained specimens.

During the testing process, various mechanical, nano-mechanical, and physical tests were conducted to examine

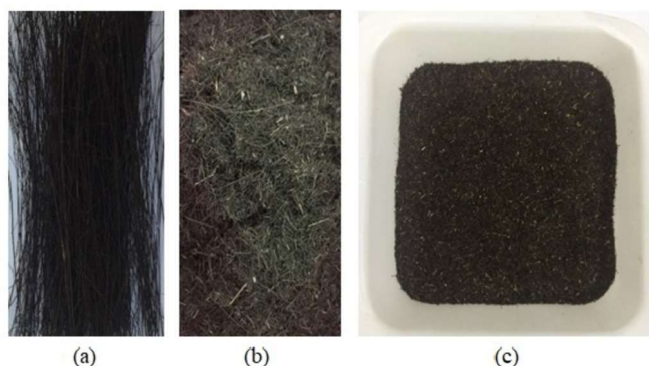
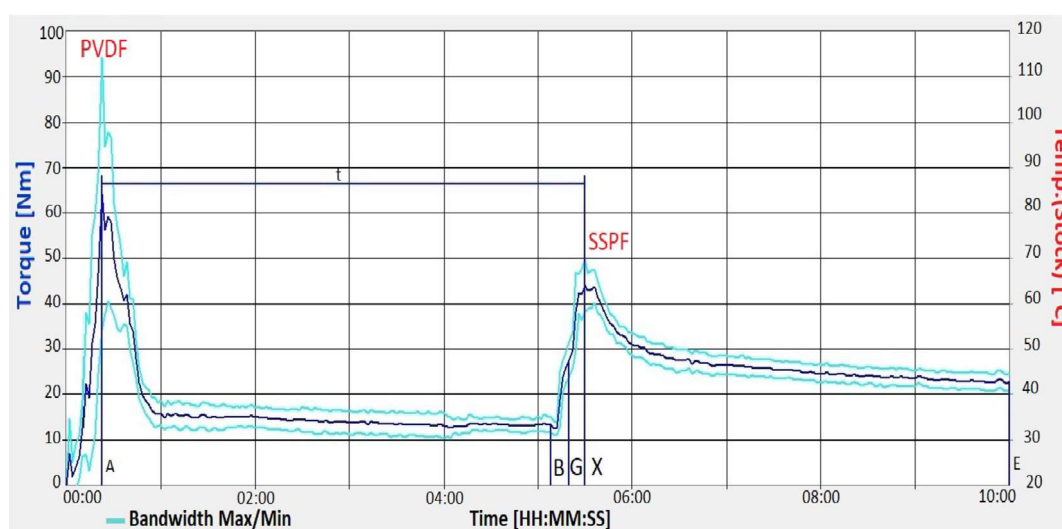
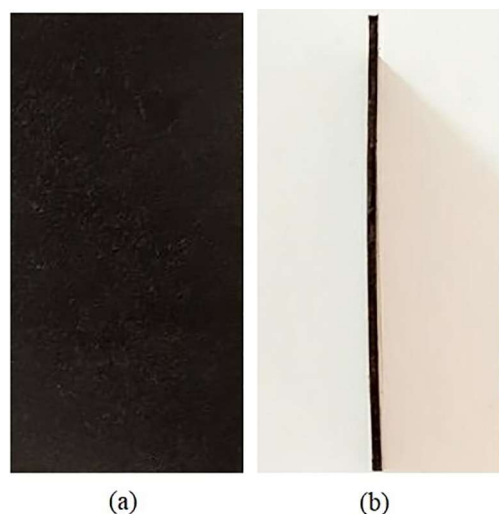


Fig. 1. a) SPF threads after the selection process; b) SPF after the crushing process; c) SSPF after the grinding process and sieved at 200–250 μm .

Table 1

The properties of PVDF used in the composites.

Thermal properties		Mechanical properties	
Melting point (°C)	169–173	Tensile modulus (Thickness 2.00 mm) (MPa)	2100–2400
Crystallization temp. (DSC peak) (°C)	134–144	Flexural modulus (Thickness 2.00 mm) (MPa)	1900–2200
Glass transition temp. T _g (°C)	–32.0	Yield strength (Thickness 2.00 mm) (MPa)	50.0–57.0
Decomposition temp. (°C) 1% wt.	375	Break strength (Thickness 2.00 mm) (MPa)	40.0–50.0
Vicat softening point (°C)	135–145	Elongation at yield (%)	5.0–10
Heat deflection temp. (°C) at 1.8 MPa (Thickness 4.00 mm), After annealing 150 °C.	105–115	Elongation at Break (%)	20–50
Electrical properties		Hardness, Shore D (Thickness 2.00 mm)	75–80
Surface resistance (ohm), Voltage <1 V, After 2min-500 V.	≥1.00E+14	Izod impact, notched (Thickness 4.00 mm, 23 °C)	60–160
Electrical resistivity (ohm * cm), Intensity = 10 mA, After 2 min at 23 °C.	≥1.00E+14	Burning properties	
Dielectric strength (Thickness 1.00 mm, 23 °C)	20–25	UL Flame retardant grade (0.100 mm)	V-0
Dielectric constant (1 kHz, 23 °C)	7.0–10.0	Oxygen index (3.00 mm)	44%
Physical properties		Molecular properties	
Density (g/cc)	1.77–1.79	Molecular weight (Da)	220,000–260,000
Melt viscosity	6–9	Intrinsic viscosity (dl/g), 30 °C.	0.8–0.9
Water absorption	≤0.040	Molecular weight distribution	1.8–2.3

**Fig. 2.** Torque-time plastogram of PVDF and SSPF blends, compounded at 175 °C and at different shear rates, to highlight fusion and degradation behaviors.**Fig. 3.** A sample of PVDF-SSPF composites; (a) Frontal view of the specimen; (b) Cross-sectional view of the specimen.

composite properties. Mechanical test consists of tensile and flexural assessment. Nanomechanical test consists of quasistatic

nanoindentation analysis. Physical test consists of density, moisture content, water absorption, and thickness swelling evaluation.

2.3.2. Mechanical testing of composites

Flexural and tensile tests were accomplished with the use of the universal testing machine Instron 5567 (INSTRON, USA). Five specimens were employed in each testing procedure. Their dimension was 150 mm (L) × 20 mm (W) × 2 mm (T) with a rate of 5.000 mm/min in the tensile test. In the flexural test, their dimension was 120 mm (L) × 13 mm (W) × 2 mm (T) with a rate of 8.5 mm/min and support span of 32 mm. A three-point bending flexural test was also conducted.

2.3.3. Nanomechanical testing

Nanomechanical testing is an important method for characterizing composites. In this work, it was used to obtain a better characterization of the PVDF-SSPF composite structure. Complex mechanical behavior can be analyzed in the smallest possible scale. Nano and micromechanical testing aid in the development of design concepts based on mechanical properties. Toughness, hardness, and fracture phenomena can be quantitatively evaluated in the new testing method. The Hysitron TI950 Triboindenter (Bruker, USA) was used for the nanomechanical test.

2.3.4. Physical testing of composites

Five specimens were prepared to determine the density of composites. Specimens were cut into small pieces (20 mm × 10 mm × 2 mm), tested at 25.4 °C water temperature, and 27 °C room temperature. The value of the density table for distilled water according to the given temperature was 0.99697. The XS205DU scaling machine (Mettler Toledo, Malaysia) was used to measure density and analytical balance. Five measurements and an average value were obtained. To determine moisture content in composites, five specimens were also prepared. Each weighed (2) g. The UniBloc moisture analyzer MOC63u (Shimadzu Corp., Japan) was used to conduct this test.

Specimens were prepared and dried for (24) h using Protech Gov-100 natural/gravity convection oven (Tech-Lab, Malaysia) for the water absorption test. Normal and prolonged immersions were accomplished. Readings were taken after (0.5) h, (1) h, (2) h, (24) h, and (100) h. Water absorption test was included in the normal and prolonged assessments. Specimens were weighed before and after immersion in distilled water and kept at room temperature of 27 °C ± 0.5. Water absorption was calculated by Eq. (1):

$$Wa = \frac{W_i - W_0}{W_0} \times 100 \% \quad (1)$$

where Wa is the water absorption, W_i is the weight of specimens after immersion in distilled water, and W_0 is the weight of specimens before immersion.

The thickness swelling test was conducted using the Digital Vernier caliper, Model: Absolute Digimatic Caliper (Mitutoyo, Japan). Three-point measurement was taken for each of the five specimens. Dimension was 20 mm (L) × 10 mm (W) × 2 mm (T). Specimens were immersed in distilled water for (100) h. Long immersion was accomplished due to insufficient water absorption of composites. Immersing specimens for a short time may not show results. An average value was obtained for each test and the thickness swelling was calculated by Eq. (2):

$$Ts = \frac{T_i - T_0}{T_0} \times 100 \% \quad (2)$$

where Ts is the thickness swelling, T_i is the weight of specimens after immersion in distilled water, and T_0 is the weight of specimens before immersion.

3. Results and discussion

3.1. Mechanical properties

Composites performed well in both tests. Tests were conducted in accordance to given parameters. The mean tensile stress at maximum load (MPa) was 23.87 with a maximum load mean (N) of 620.72, load at break (N) of 599.54, and energy at break (J) of 0.64. Fig. 4 displays the tensile stress versus tensile strain curves of PVDF-SSPF specimens.

The five specimens displayed similar performance. The variation between the highest and lowest tensile stress at maximum load (MPa) was 5. Variation of the maximum load (N) between the highest and lowest specimens was 146. This indicates high consistency in the composites. The mean of the maximum flexural stress (MPa) was 52, the modulus (MPa) was 2151, and the extension at break (mm) was −17. Fig. 5 presents the flexural stress versus the flexural strain curves of PVDF-SSPF specimens.

The flexural strength and flexural modulus are determined. For the maximum flexure stress, the variation between the highest and lowest specimens was 18 (MPa). The maximum flexure stress

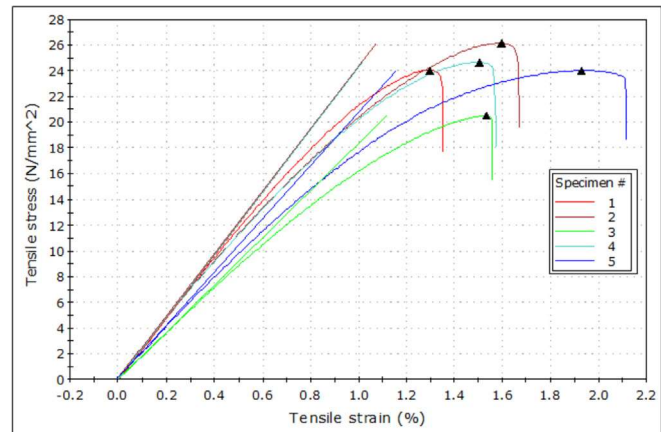


Fig. 4. Tensile stress versus tensile strain curves of PVDF-SSPF specimens.

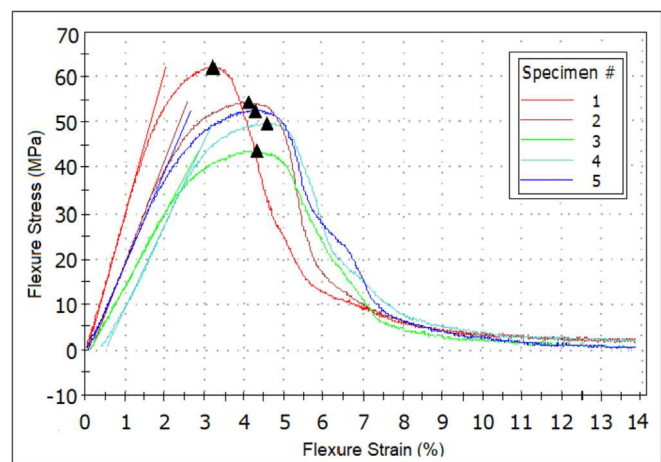


Fig. 5. Flexural stress versus flexural strain curves of PVDF-SSPF specimens.

averaged at 52 (MPa) and the flexural modulus averaged at 2151 (MPa). The overall tensile and flexural results exhibited favorable properties. The combination of the fiber's strength properties and interfacial bonding in the fiber-matrix seems to prominently contribute to the final flexural strength of the PVDF-SSPF composites. Tables 2 and 3 provide further details on the results.

The fiber's tensile properties are enhanced when the PVDF polymer is incorporated. The tensile stress at break was 23.059 (MPa), and the stress at maximum load was 23.874 (MPa). The applied maximum load was 620.729 and the load at break was 599.546. The average of Young's modulus and modulus (Aut.) were 2243.799 (Mpa) and 2204.015 (Mpa). Enhancements in tensile properties are due to favorable characteristics of both the fiber and matrix. The flexural test also demonstrated the composites' exceptional properties. Table 3 provides further information.

Table 3 provides the average values and standard deviation of the five specimens. The maximum flexure stress was 52.493 (MPa). This is relatively good compared to the highest result which was 61.896 (MPa). The flexure load at break and stress at break were 1.915 (N) and 1.768 (MPa). The flexure load at maximum flexure stress was 56.867 (N). The modulus was 2151.159 (MPa) in the flexural test. Flexural properties displayed excellent results when compared to raw properties and other composite structures. The breakage points of tensile and flexural specimens were scanned using scanning electron microscopy (SEM) machine, EM-30AX. The surface morphology of specimens was investigated (shown in

Table 2
Results obtained in the tensile strength test.

Tensile testing					
	Tensile stress at Break (Sta.) (MPa)	Tensile strain at Break (Sta.) (mm/mm)	T. stress at Max Load (MPa)	Modulus (Aut. Young's) (MPa)	Modulus (Aut.) (MPa)
1	22.73070	0.01348	>24.04494	2449.90011	2441.22385
2	25.00709	0.01664	>26.15460	2430.48371	2366.76444
3	20.21652	0.01556	>20.50727	1839.70387	1835.52525
4	24.03486	0.01564	>24.67086	2426.91546	2368.52605
5	23.30833	0.02114	>23.99347	2071.99305	2008.03821
Mean	23.05950	0.01649	> 23.87423	2243.79924	2204.01556
Standard Deviation	1.80257	0.00284	2.07452	275.53160	266.46337
	Maximum Load (N)	Tenacity at Break (Sta.) (gf/tex)	Energy at Break (Sta.) (J)	Load at Break (Sta.) (N)	Tensile extension at Break (Sta.) (mm)
1	625.16844	602.65034	0.50465	590.99810	1.34775
2	680.01967	663.00359	0.70251	650.18440	1.66403
3	533.18895	535.99303	0.49665	525.62959	1.55641
4	641.44241	637.22704	0.62391	624.90625	1.56423
5	623.83035	617.96494	0.88958	606.01657	2.11403
Mean	620.72996	611.36779	0.64346	599.54698	1.64929
Standard Deviation	53.93750	47.79076	0.16222	46.86672	0.28416

Table 3
Results obtained in the flexural test.

Flexural testing					
	Maximum Flexure stress (MPa)	Flexure load at Break (Sta.) (N)	Flexure strain at Break (Sta.) (mm/mm)	Flexure stress at Break (Sta.) (MPa)	Extension at Break (Sta.) (mm)
1	61.89687	1.99547	0.20097	1.84197	–17.14906
2	54.37067	1.56499	0.20215	1.44461	–17.24983
3	43.73454	0.70026	0.20259	0.64640	–17.28738
4	49.85807	3.40142	0.20268	3.13977	–17.29505
5	52.60615	1.91473	0.20213	1.76744	–17.24812
Mean	52.49326	1.91537	0.20210	1.76804	–17.24589
Standard Deviation	6.62618	0.97647	0.00068	0.90136	0.05817
	Modulus (Aut.) (MPa)	Flexure strain at Maximum Flexure stress (mm/mm)	Flexure load at Maximum Flexure stress (N)	Load at Break (Standard) (N)	Load at Maximum Flexure stress (N)
1	3066.61552	0.03240	67.05495	–1.99547	–67.05495
2	2216.43434	0.04106	58.90156	–1.56499	–58.90156
3	1589.41833	0.04341	47.37908	–0.70026	–47.37908
4	1845.82577	0.04590	54.01291	–3.40142	–54.01291
5	2037.50270	0.04269	56.98999	–1.91473	–56.98999
Mean	2151.15933	0.04109	56.86770	–1.91537	–56.86770
Standard Deviation	562.14327	0.00516	7.17836	0.97647	7.17836

Fig. 6).

Under increasing tensile and flexural stresses, the matrix composite gradually encounters a transition. This yields various failure modes such as fiber-matrix debonding, fiber pullout incidents, fiber breakage, matrix cracking, and matrix yielding. When the fiber-matrix interaction force is exceeded by the shear force where the load is transferred, a debonding results. When the stress level exceeds the fiber stress, fiber fracture will occur. This results in fiber pullout and fracture incidents (Bachtiar et al., 2009). Fiber-matrix debonding may also occur due to adhesion disruption or fiber disarrangement at the interface. Cracks are determined by identifying particular regions adjacent to the fiber where it is less dense or where regression occurred due to enormous stress applied. The fractured SSPF can be seen within the matrix composites. The interfacial region between the fiber and matrix shows good mechanical cohesion. The fiber is likely to be ruptured and fractured in the tensile test. It was fractured in the flexural test and pullout cases were noted. The fiber appears to be strong and has excellent

merging properties with the matrix composites.

3.2. Nanomechanical properties

The fiber-matrix interphase has a fundamental influence on the load-transfer process. This influences most mechanical properties of composites. Their singular phases (the fiber, matrix, and interphase) are based on their mechanical interaction. The quantitative measurement of the mechanical performance at or near the fiber-matrix interphase is of paramount importance (Zhao et al., 2009). Natural fiber composites have been criticized for the poor relationship between the hydrophilic nature of fibers and hydrophobic polymers (Bachtiar et al., 2010). Chemical compatibility is an important aspect in controlling adhesion and dispersion between the fiber and fillers. The hydrophilic surface of cellulose resulted in poor interaction (Reddy et al., 2013). Characterizing nanomechanical properties of natural fiber composites can improve fiber-matrix compatibility and interfacial adhesion. The synthesis,

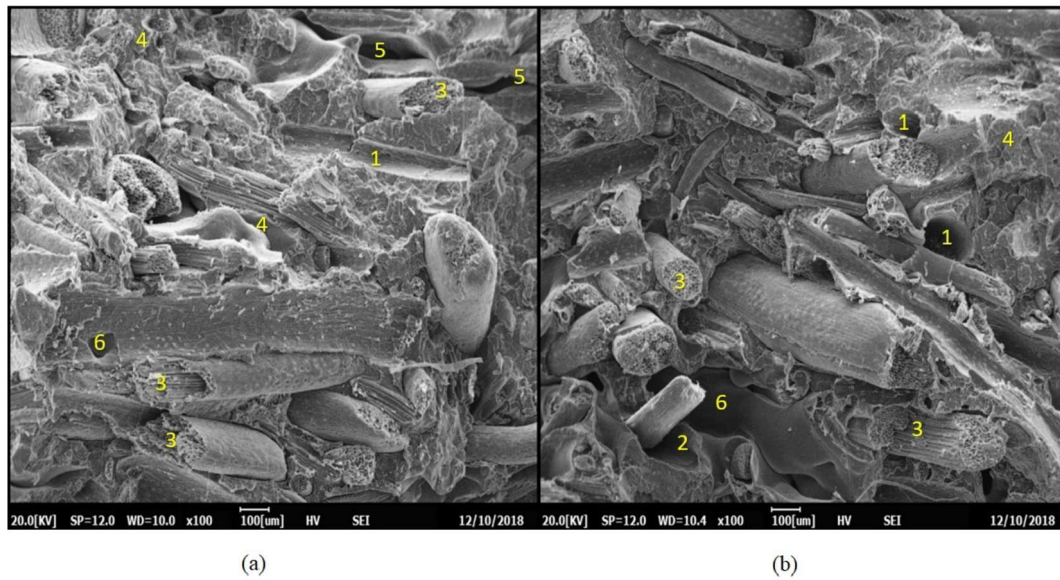


Fig. 6. Surface morphology from scanning electron microscopy photographs of PVDF-SSPF fractured surfaces; (a) Fracture surface of tensile test; (b) Fracture surface of flexural strength test; (1) Fiber pullout; (2) Fiber-matrix debonding; (3) Fiber breakage; (4) Matrix yielding; (5) Matrix cracking; (6) Void.

morphology, topography, and modulus mapping of PVDF-SSPF composites are presented in this section. The quasistatic nano-indentation was conducted to quantify hardness, Young's modulus, and fracture phenomena. Fig. 7 provides the scan-line analyses of the nanomechanical testing of PVDF-SSPF composites. The hardness and modulus under the same nano specimen for two random surficial regions are displayed.

The scan line plot provides real-time analysis on the hardness and modulus properties of two selected regions (see Fig. 8). The scan line is in the form of a histogram plot to provide an accurate evaluation of the nanoscale measurement. The counts represent the number of accomplished scans of the current testing process. The probe either completes the task or is disengaged from the surface. The two regions exhibited different results. There is relative

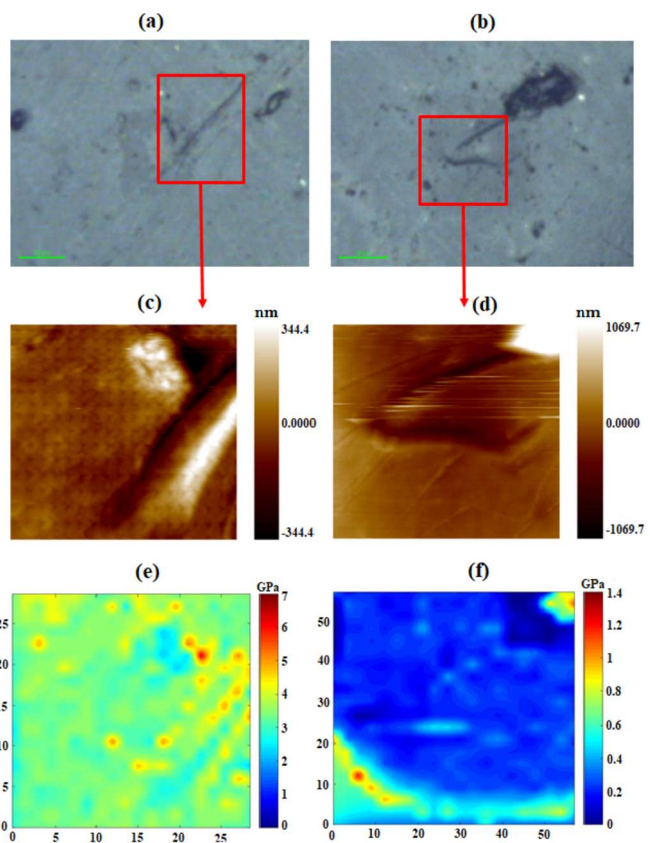


Fig. 8. Surface morphology, in-situ SPM imaging, and modulus mapping of PVDF-SSPF composites; (a) & (b) 20 & 30 μm incorporated fiber into matrix; (c) & (d) Close-up image of nm showing the same perspective; (e) & (f) Dynamic modulus mapping of the same captured area.

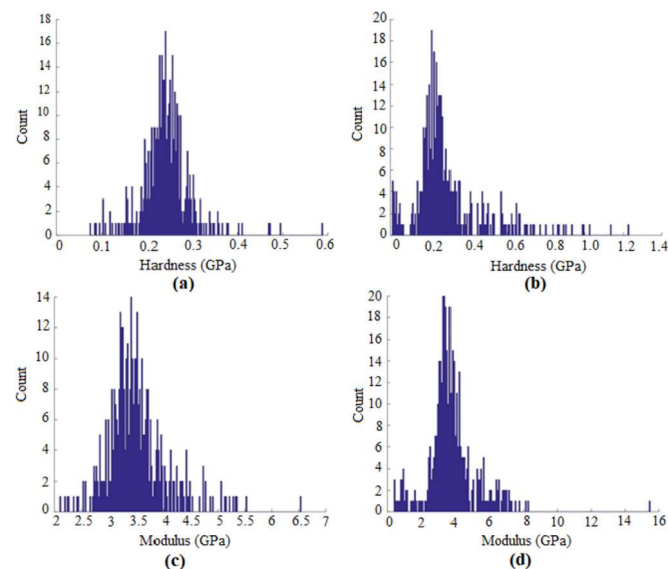


Fig. 7. Scan line analyses of nanomechanical testing of PVDF-SSPF; (a) & (b) Two hardness analyses on surficial regions; (c) & (d) Two modulus analyses on surficial regions.

similarity in hardness properties. Larger differences are found in modulus properties. Both results of hardness and modulus (GPa) exhibited good composite properties. The results provide a range of hardness and modulus values for each test. This indicates a

homogeneous distribution of fiber in the matrix. Slight variance in density may have caused the widened range of hardness and modulus. This can be due to a very slight variance in the morphological nature of the composite. The fiber and matrix may not have maintained a perfect flattened surface area. From Fig. 7, the hardness and modulus for the two tests displayed favorable consistency. The ultimate average reveals good nanoindentation modulus and hardness phenomena in the composites. This is essential for analyzing and predicting their mechanical behavior for advanced applications. Interpreting mechanical behavior will introduce novel methods for simultaneously predicting durability, hardness, modulus, fatigue life, and adhesion properties. The indentation hardness and modulus focus on the interfacial zone around the fiber and fiber-matrix. This provides modulus mapping to better understand the role of each element within the composite. Substantial nano-bonding in the fiber-matrix was observed. The incorporation of fiber threads significantly contributed to the overall strength of the composite. Similar improvements were maintained in previous studies (Lee et al., 2008). Improvements of mechanical properties with the presence of compatibilizers and fiber surface treatment were reported (Arbelaiz et al., 2005). Increase in the area of contact between the matrix and fiber will improve the level of adhesion. Incorporating a mechanical adhesion component may be required (Herrera-Franco and Valadez-González, 2005). Mechanical improvements were observed in short fiber. No significant result was observed when the length of fiber increased. Fiber length increase may show a drop in tensile strength and tensile modulus. Composites that include short fiber are subjected to higher levels of dimensional changes and water absorption. The hybridization and incorporation of fiber within matrix composites have exhibited positive effect when optimizing the relative volume fraction. Improved modulus properties are obtained in modified hybrid composites (Jawaid and Abdul Khalil, 2011). Fig. 8 provides further information.

Recent improvements on nanoindentation techniques opened further prospects to characterize the mechanical properties of composites and heterogeneous polymers at micrometer scales. Nanoindentation is a powerful tool that determines the mechanical properties of fibers and identify the influence of structure, treatment, damage, etc. (Bourmaud and Baley, 2009). Direct methods for investigating interphase characteristics include modulus, stiffness, and hardness. Interphase mechanical properties are different from bulk materials. Dimensional information can be obtained from nanomechanical testing as an alternative (Gu et al., 2010). Modulus analysis provides novel and unprecedented characterization of composites. Two different areas of PVDF-SSPF composites were further investigated. The segments with existing short fiber exhibited better stiffness. Various modulus values with good reinforcement properties were acquired. The correlation between fiber morphology and high values of Young's modulus is apparent. The fiber induces the mechanical compound of the matrix composite. Lower values may exist in micro lumen areas between the “fibers” on the surface morphology or from varying modulus results that occur from a change of load. It is imperative to consider the aspect ratio when determining fiber length. The aspect ratio is the most influential parameter that impacts the mechanical properties of short fiber reinforced polymer. Enhancing geometric features can lead to outstanding reinforcement properties (Bourmaud and Pimbert, 2008).

3.3. Physical properties

Tests conducted on the physical properties of PVDF-SSPF are crucial for recommending applications of the composite. Density test was conducted to determine the intensive property and effect

of fiber loading (Kasim et al., 2016). A moisture content test was accomplished to identify the mechanical behavior of composites (Mizera et al., 2017). Water absorption test (Munoz and García-Manrique, 2015) and thickness swelling (Rozman et al., 2000) were further conducted to understand the dimensional properties, distribution of fiber, and mechanical properties of composites.

3.3.1. Density

Density and weight values of PVDF-SSPF specimens are displayed in Fig. 9. Natural fibers are known for lower density compared to inorganic fibers used as reinforcements (Bachtia et al., 2008). Lower density is a favorable attribute (Oumer and Bachtia, 2014). SPF is known for its low density compared to other fibers. SPF displayed enhanced mechanical and physical properties such as hardness, tensile strength, tensile modulus, density, and moisture behavior. Overall density of the five specimens was 1.58 g cm^{-3} . The incorporation of SSPF led to the reduction of composite density. An increase in fiber content can cause a slight decrease in the overall density of the composite (Jumaidin et al., 2017). Low density of SPF leads to good physical and mechanical properties which may even outrank those of glass fibers (Sanyang et al., 2016).

Minor decrease in density is expected and observed in the given analysis when SPF was added to the composite. The lowest density value was for SP1 with 1.52 g cm^{-3} . The highest values were observed in SP3 and SP4 with 1.62 g cm^{-3} . Density reduction can be due to the fibrous nature of SPF which maintains lower density when compared to solid materials. Similar results were also reported in previous works (Jumaidin et al., 2017). Lower density may occur when treating or removing certain components of the fiber. Increased volume and weight loss will cause the density value to decrease (Ilyas et al., 2017). Density decreases with fiber content. The composites become more suitable and attractive for lightweight applications (Ramanaiah et al., 2012). Other studies reported an increase in density when adding fiber. This can be due to the close packing of fiber within the composites (Kasim et al., 2016).

3.3.2. Moisture content

Fig. 10 provides information on the moisture content analysis. The PVDF-SSPF composites displayed very little moisture content. This is due to the homogenous incorporation of fibers and polymer, enhanced interfacial bonding between the matrix and fibers, and absorption hindrance caused by fibers (Sahari et al., 2013).

The average moisture content reported is 0.24%. SPF is known for its low moisture content which improves the physical

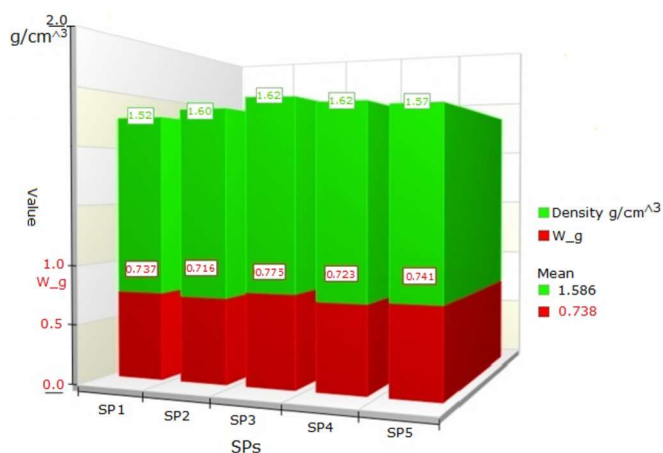


Fig. 9. Density-weight values of PVDF-SSPF composites.

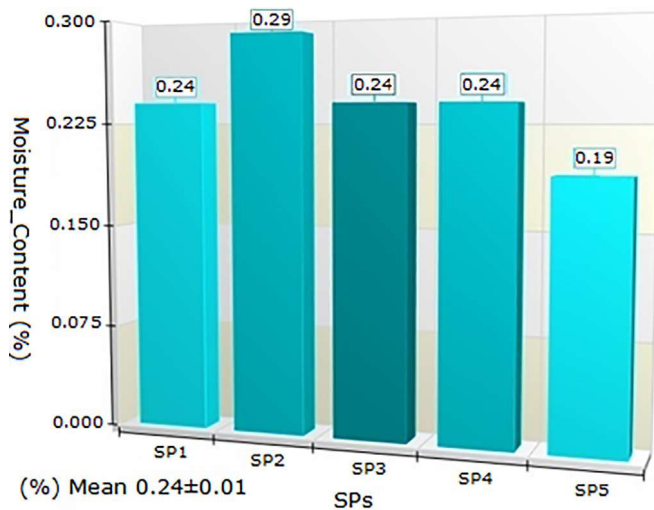


Fig. 10. Moisture content of PVDF-SSPF composites.

properties of composites (Edhirej et al., 2017). Microcracking and charging can be caused by high moisture content in fibers employed in composite materials (Leman et al., 2008a). Moisture absorption behavior is classified as the equilibrium moisture content and duration of the transient. The first classification characterizes the composite's affinity for hydrophilicity. The second is related to the thickness of the sample and the diffusion coefficient (diffusivity). This reflects the transport rate of water molecules within the composite (Leman et al., 2008b). Lower moisture absorption and moisture content qualify composites to a wider range of applications. Low moisture resistance affects the long-term properties of composites. Research is needed to extend their application range by improving moisture resistance and fire retardance (Pickering et al., 2016). Moisture absorption is one of the major concerns in outdoor applications. It affects long-term

performance and has adverse effects on properties. The use of compatibilizers, coupling agents, or chemical modifications can improve the moisture resistance of these composites (Wang et al., 2006).

3.3.3. Water absorption

A gradual water absorption test was accomplished in five stages, from 0.5 h to 100 h. The test provided an extensive view on how composites behave in water, revealing accurate information on the sensitivity of bio-composite materials when exposed to water. Weak interfacial adhesion between the fiber and matrix may cause an increase in water absorption. This is due to the emergence of hydrogen chemical bonds between water molecules and cellulose fiber (Munoz and García-Manrique, 2015). A decrease in water absorption or uptake was observed with increasing fiber loading. This could be attributed to the hydrophobic behavior of SPF (Sahari et al., 2013). The incorporation of SPF improves composites' physical properties. It decreases water absorption, solubility, and water content while increasing relative crystallinity (Edhirej et al., 2017). (See Table 4)

Lower uptake of water can be justified by the following: better interfacial bonding between the fiber and matrix, possible reduction of void content in composites, hindered diffusion of water due to higher fiber content, and greater affinity of the matrix for water compared to fibers (Ramírez et al., 2011).

3.3.4. Thickness swelling

Table 5 provides a summary of obtained results. Composites exhibited less thickness swelling when immersed in distilled water for 100 h. This signifies their excellent resistance to swelling conditions. SPF exhibited less hydrophilic behavior with a better rigid structure. The cumulative pore volume was closely aligned. The fiber maintained firm and tightly bound building elements via robust hydrogen-bonded cellulose structure. This left insignificant interfacial voids (Ilyas et al., 2018). The hydrophilic attribute of natural fiber is usually the determinant for water absorption. It influences the thickness swelling. The dimensions of lignocellulosic

Table 4
Summary of water absorption test results of PVDF-SSPF composites.

SPs	Weight/g	0.5 h	Water Absorption (%)	1 h	Water Absorption (%)
SP1	0.731	0.733	0.27359781	0.733	0.27359781
SP2	0.712	0.713	0.14044944	0.713	0.14044944
SP3	0.770	0.771	0.12987013	0.771	0.12987013
SP4	0.719	0.720	0.13908206	0.720	0.13908206
SP5	0.737	0.738	0.13568521	0.738	0.13568521
*Mean	0.7338	0.735	0.1635323	0.735	0.1635323
2h	Water Absorption (%)	24h	Water Absorption (%)	100h	Water Absorption (%)
0.734	0.41039672	0.739	1.09439124	0.752	2.872777
0.714	0.28089888	0.716	0.56179775	0.719	0.983146
0.772	0.25974026	0.774	0.51948052	0.778	1.038961
0.721	0.27816412	0.724	0.69541029	0.728	1.251739
0.738	0.13568521	0.741	0.54274084	0.747	1.356852
0.7358	0.27255383	0.7388	0.68138457	0.744	1.499046

Table 5
Summary of thickness swelling test results for PVDF-SSPF composites.

SPs	Room Temp. °C	Water Temp. °C	3-points Averaged Dimensions (mm)	3-points Averaged Swellings (mm)	100 h Thickness swelling (%)
SP1	27 ± 0.5	25.4 ± 0.01	2.09	2.15	2.86 ± 0.01
SP2	27 ± 0.5	25.4 ± 0.01	2.14	2.19	2.64 ± 0.01
SP3	27 ± 0.5	25.4 ± 0.01	2.01	2.05	1.98 ± 0.01
SP4	27 ± 0.5	25.4 ± 0.01	2.05	2.11	2.75 ± 0.01
SP5	27 ± 0.5	25.4 ± 0.01	1.98	2.02	2.18 ± 0.01
*Mean			2.05	2.10	2.48 ± 0.01

materials change with different moisture content. The capillary action and hydrophilic properties of lignocellulosic materials cause water intake when soaked. They increase the dimensions of the composite (Jawaid et al., 2011). Poor resistance to water absorption occurs in lignocellulosic materials due to the existence of polar groups. This entices water molecules through the bonding of hydrogen and causes moisture to build-up in the fiber's cell wall and fiber-matrix interface. Instability and alterations occur in the dimensions of cellulose-based composites. Changes will be observed in their thickness and linear expansion (Ashori and Sheshmani, 2010). An improvement of composites' dimensional stability was seen when SSPF was incorporated. The reduction in water uptake yielded better interfacial bonding and higher resistance to water absorption.

4. Conclusions

PVDF-SSPF composites were fabricated and assessed under various mechanical, nanomechanical, and physical tests. Their mechanical and physical characteristics make them excellent materials for outdoor and light applications. The composites revealed high compatibility with the fiber-matrix. The utilization of short fiber contributed to interfacial bonding with stress-transfer efficiency. This is due to the homogenous composite structure and substantially improved interfacial matrix-fiber bonding. The quasi-static nanoindentation provided distinctive results on the nanomechanical properties of the composites. Nanomechanical testing revealed good compatibility of fiber behavior. It provided a correlation between the fiber and its morphology and high values of Young's modulus. The contribution of fiber demonstrated strength, hardness, Young's modulus, stiffness, and interfacial phenomena. The composites exhibited outstanding physical properties with excellent resistance to water absorption and thickness swelling. Their substantial attributes can be configured for advanced applications such as photovoltaic backsheets, interior parts in automotive industry, decorative designs, outdoor products, and eco-design products. Further efforts are needed to better allocate these composites based on their intended functionality. This may include investigating their optical, thermal, and technical properties. The composites can enhance the sustainability and productivity of green production and minimize hazardous waste.

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